seismoSQL, a practice in using **SQL Server** commands in field of Seismology

Konun Koushesh - 17 Okt. 2025

It starts with generating 6 tables of data as explained shortly in the next paragraph and follows by applying/defining **procedure**, **trigger**, **index**, **function** and **view** on the tables of data.

Tables of data

In seismology sensors are installed to record vibrations. Vibrations can be generated from any type of sources. Using physics laws location of the sources are found by seismologists. Mapping those locations gives worthfull information about the active underground geological structures such as faults and/or channels transporting magma in volcanic regions.

Sensors are installed in places with private or public ownerships and could be installed by different institutions. Such infos inserted in the following tables:

t_stations
t_ownerStationSite
t_stInstalService
*(sensors = stations)

Sensors are serviced regularly. It is common that some sensors are removed or some added to the existing or new locations in time.

Vibrating sources (events) used in this practice are small earthquakes occurred due to magma transportation under the east Eifel volcanic field (close to Laacher See volcano), Germany (Konun Koushesh 2025, https://publikationen.bibliothek.kit.edu/1000180160). They are summarised in the following tables:

t_events
t_eventsQuality
*(event = source = earthquake).

Just for making a scenario to more practice with SQL commands, location of nearby villages and cities up to 1000 km from the Laacher See volcano is stored in table:

t_remarkablePlaces

The first three rows in this table present three assuming eruption points.

One scenario is that which villages/cities get affected by a volcanic eruption earlier than the other ones (sorted by distance to each of the 3 assuming eruption points). Results of using SQL commands are restored in the following tables:

t_remarkablePlacesCloseToeruptionCenter1,
t_remarkablePlacesCloseToeruptionCenter2,
t_remarkablePlacesCloseToeruptionCenter3.

Other scenario is related to doing services on the installed sensors and restoring the changes in a new table called

t_stInstalService_delet_update

and the rests is doing some common queries about the table of data.

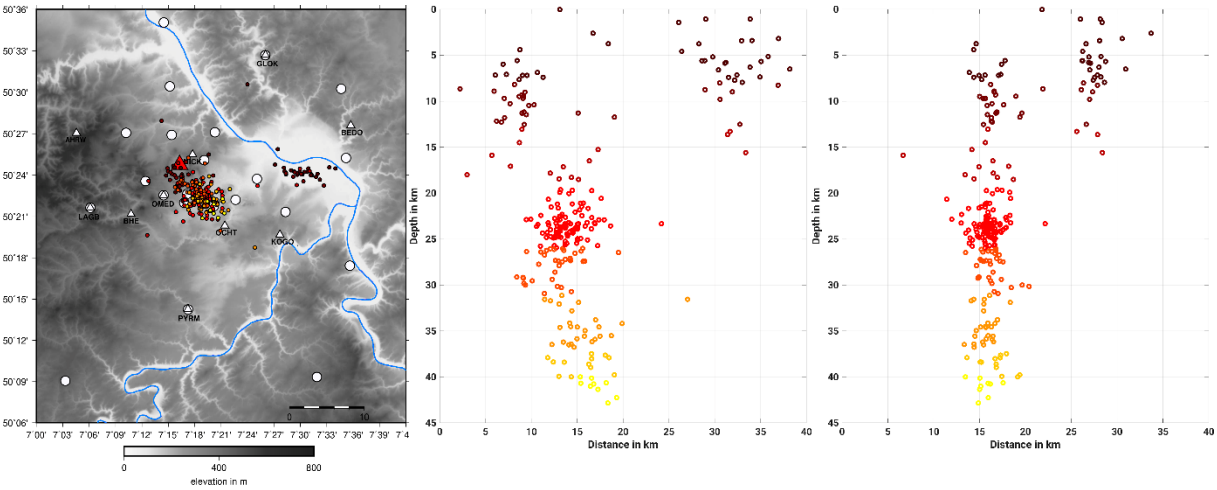

Figure 1, shows active sources in magma transportation in the east Eifel volcanic field, Germany. The triangle (location of Laacher See volcano), the centers of the main cluster of events (close to the triangle) and the east (small) cluster of events are assumed as the possible eruption points in a hazard assessment scenario in this work.

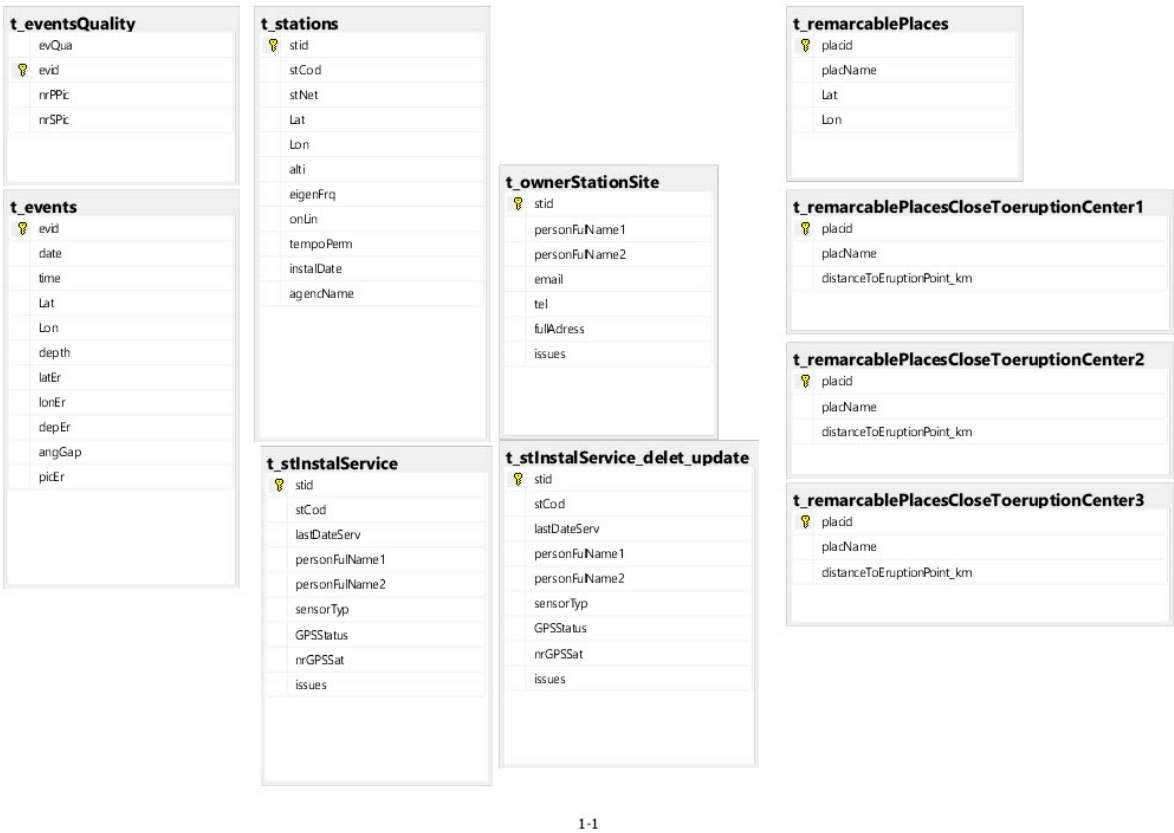

Figure 2, overview of the initial tables and the tables are generated during the practice on SQL Server commands.

Short explanation on what is done on data of tables:

Procedure: Joins date of two tables: **t_stations** and **t_ownerStationSite**

Trigger: writes changes (such as inserting, updating, and deleting) done in **t_stInstalService** into **t_stInstalService_delet_update**

Index: An index command to speed up the search (filter) when the goal is separating main and east cluster of events in **t_events** table.

This is followed by some small queries in the **t_events** table and then some updates in the **t_remarkablePlaces** table.

Function: Takes a location ID from the table **t_remarkablePlaces** and returns the latitude and longitude of the location. There is also a function that calculates the distances between each location in the same table and the assumed eruption points.

View: showing the content of the tables. There are also views that combine data from the two tables **t_events** and **t_eventsQuality**, and another view that combines data from the three tables **t_stations**, **t_stInstalService** and **t_ownerStationSite**.

Scripts plus expanded version of explanation (including 8 section):

1. Creating Tables of data

```
--+ for events (earthquakes) and their properties
CREATE TABLE t_events(
      --evQua char(1) Null, -- quality of picking on event
      evid int IDENTITY(1,1) PRIMARY KEY NOT NULL, -- event ID
      date date NULL, -- date of event occurrence
      time time(3) NULL, -- origin time of event occurrence
      Lat float NULL, -- Latitude of event
      Lon float NULL, -- Longitude of event
      depth float NULL, -- depth of event
      latEr float NULL, -- location error in Lat. direction
      lonEr float NULL, -- location error in Lon. direction
      depEr float NULL, -- location error in depth direction
      angGap int null, -- angular gap while picking and localization process
      picEr float NULL, -- average time error in s in picking phases
      --FOREIGN KEY (evQua) REFERENCES t_eventQuality(evQua)
)
GO
-- drop table t_events

--+ for additional info on events quality
CREATE TABLE t_eventsQuality(
      evQua char(1) NOT NULL,  -- quality of picking on event
      evid int identity(1,1) PRIMARY KEY  not null, -- event ID
      nrPPic int null, -- number of picks of P-phase arrivals
      nrSPic int null -- number of picks of S-phase arrivals
      --FOREIGN KEY (evid) REFERENCES t_events(evid)
)
GO
-- drop table t_eventsQuality

--+ for stations' properties
CREATE TABLE t_stations(
      stid int IDENTITY(1,1) PRIMARY KEY NOT NULL, -- station ID
      stCod char(4) NULL, -- station code known for the seismological networks
      stNet char(2) NULL, -- seismological network that ech station belongs to
      Lat float Null, -- latitude of the location in which the sensor is installed
      Lon float NUll, -- longitude of the location in which the sensor is
installed
      alti float NULL, --elevation (altitude) of the location in which the sensor
is installed
      eigenFrq float NULL, -- eigen frequency of the sensor
      onLin nvarchar(3) NULL, -- data transfering status online or offline
      tempoPerm nvarchar(10) NULL, -- temporary or permanent location for station
      instalDate date NULL, -- date of installation
      agencName nvarchar(50) NULL -- the supporting agency
)
GO
-- drop table t_stations

--+ for installations and services of the sensors (stations)
CREATE TABLE t_stInstalService(
      stid int IDENTITY(1,1) PRIMARY KEY NOT NULL, -- station ID
      stCod char(4) NULL,-- station code
      lastDateServ date NULL, -- latest service date
      personFulName1 nvarchar(50) NULL, -- the person who installed the sensor
      personFulName2 nvarchar(50) NULL, -- the assistant person while installation
      sensorTyp nvarchar(30) null, -- sensor name (type)
      GPSStatus nvarchar(3) null, -- if GPS works correctly
      nrGPSSat int null, -- how many satellites are available at the time of
installation
```

```
	issues nvarchar(50) default 'no issues'
)
GO
-- drop table t_stInstalService

--+ for owner of the sites that sensors are installed there and their contact
infos
CREATE TABLE t_ownerStationSite(
	stid int IDENTITY(1,1) PRIMARY KEY NOT NULL, -- station ID
	personFulName1 nvarchar(50) NULL, -- the first contact person
	personFulName2 nvarchar(50) NULL, -- the second contact person
	email nvarchar(50) NULL, -- email
	tel nvarchar(30) NULL, -- telephon
	fullAdress nvarchar(50) NULL, -- address to pick up the key of the site
	issues nvarchar(50) default 'available on working days 8:00 - 16:00' -- any
remarks write here
)
GO
-- drop table t_ownerStationSite

--+ for remarkable places (locations) around the east Eifel volcanic fields
(Laacher See volcano Germany)
CREATE TABLE t_remarkablePlaces(
	placid int IDENTITY(1,1) PRIMARY KEY NOT NULL, -- place ID
	placName nvarchar(50) NULL, -- name of village/city/location
	Lat float NULL, -- latitude
	Lon float NULL, -- longitude
)
GO
-- drop table t_remarkablePlaces

--+ for remarkable places (locations) around the east Eifel volcanic fields
(Laacher See volcano Germany):
-- plus their distace to eruption center 1
CREATE TABLE t_remarkablePlacesCloseToeruptionCenter1(
	placid int PRIMARY KEY NOT NULL, -- place ID
	placName nvarchar(50) NULL, -- name of village/city/location
	distanceToEruptionPoint_km float NULL, -- latitude
)
GO
-- drop table t_remarkablePlacesCloseToeruptionCenter1

-- plus their distace to eruption center 2
CREATE TABLE t_remarkablePlacesCloseToeruptionCenter2(
	placid int PRIMARY KEY NOT NULL, -- place ID
	placName nvarchar(50) NULL, -- name of village/city/location
	distanceToEruptionPoint_km float NULL, -- latitude
)
GO
-- drop table t_remarkablePlacesCloseToeruptionCenter2

-- plus their distace to eruption center 3
CREATE TABLE t_remarkablePlacesCloseToeruptionCenter3(
	placid int PRIMARY KEY NOT NULL, -- place ID
	placName nvarchar(50) NULL, -- name of village/city/location
	distanceToEruptionPoint_km float NULL, -- latitude
)
GO


-- drop table t_remarkablePlacesCloseToeruptionCenter3
```

2. Inserting data into the tables

```
insert into t_events(date,time,lat,Lon,depth,latEr,lonEr,depEr,angGap,picEr)
values
('2015-04-14', '21:23:45.055', 50.3962, 7.2858, 11.010, 2.3, 1.6, 2.6, 75, 0.36),
('2015-04-14', '21:25:53.705', 50.3948, 7.2703, 11.830, 2.4, 1.9, 3.7, 76, 0.19),
('2015-04-14', '21:26:53.384', 50.4003, 7.2546, 9.136, 2.3, 1.6, 3.6, 85, 0.21),
('2015-04-14', '21:28:33.247', 50.4003, 7.2735, 8.912, 2.1, 1.4, 3.4, 68, 0.32),
('2015-09-12', '01:31:03.126', 50.4047, 7.4673, 3.790, 2.6, 2.0, 5.0, 92, 0.11),
('2016-04-26', '20:25:49.514', 50.3872, 7.5223, 6.230, 3.0, 2.5, 4.6, 83, 0.18),
('2016-04-28', '19:09:58.505', 50.3913, 7.4954, 5.843, 3.7, 2.3, 4.8, 177, 0.14),
('2016-05-18', '21:16:46.799', 50.3541, 7.3311, 42.820, 3.9, 3.9, 4.1, 86, 0.16),
('2016-06-23', '15:14:34.200', 50.3714, 7.3314, 35.711, 2.4, 1.8, 3.8, 48, 0.18),
('2016-06-23', '17:58:59.173', 50.3709, 7.2882, 33.948, 4.0, 2.6, 4.5, 64, 0.19),
('2016-06-24', '17:55:25.513', 50.3792, 7.2946, 30.691, 2.3, 2.0, 2.7, 50, 0.12),
('2016-12-20', '18:28:11.719', 50.3534, 7.3002, 40.090, 3.7, 3.3, 4.3, 87, 0.21),
('2017-05-17', '03:48:51.043', 50.3669, 7.3455, 37.750, 3.9, 2.3, 4.3, 133, 0.14),
('2017-06-04', '20:36:32.951', 50.3751, 7.2996, 23.728, 2.0, 2.6, 3.7, 92, 0.07),
('2017-06-06', '19:22:02.357', 50.3710, 7.3160, 23.401, 3.1, 1.7, 3.4, 95, 0.16),
('2017-06-06', '19:23:03.145', 50.3713, 7.3053, 24.472, 2.9, 2.3, 3.8, 103, 0.14),
('2017-06-06', '20:54:06.609', 50.3685, 7.2784, 24.255, 3.1, 2.5, 4.3, 87, 0.16),
('2017-06-07', '10:55:49.214', 50.3842, 7.3031, 26.086, 2.0, 2.1, 2.2, 54, 0.29),
('2017-06-07', '11:02:29.275', 50.3810, 7.2977, 26.046, 2.8, 1.9, 2.9, 56, 0.16),
('2017-06-07', '11:07:35.929', 50.3757, 7.3257, 27.400, 3.5, 2.3, 2.3, 107, 0.16),
('2017-06-07', '11:08:33.781', 50.3788, 7.2844, 23.437, 2.0, 1.6, 4.2, 51, 0.36),
('2017-06-07', '11:35:24.297', 50.3746, 7.3057, 24.560, 1.8, 1.8, 3.7, 52, 0.19),
('2017-06-07', '20:59:17.039', 50.3749, 7.3089, 23.571, 2.1, 1.5, 3.1, 52, 0.16),
('2017-06-07', '21:18:29.779', 50.3820, 7.3191, 24.511, 2.3, 1.7, 3.3, 51, 0.29),
('2017-06-09', '00:43:02.879', 50.3762, 7.2959, 23.624, 2.7, 2.1, 3.8, 66, 0.17),
('2017-06-09', '02:07:50.383', 50.3735, 7.3066, 23.332, 2.7, 1.9, 3.4, 57, 0.09),
('2018-06-05', '22:34:41.988', 50.3729, 7.3123, 37.478, 2.5, 1.8, 3.6, 65, 0.20),
('2018-06-30', '03:29:12.984', 50.3731, 7.3126, 36.595, 2.6, 1.8, 4.6, 58, 0.09),
('2018-10-05', '01:27:56.479', 50.3996, 7.2716, 9.030, 1.9, 2.6, 3.7, 54, 0.10),
('2019-03-22', '21:35:09.702', 50.3892, 7.5171, 6.065, 3.2, 1.9, 4.8, 170, 0.21),
('2019-03-23', '03:09:37.144', 50.3934, 7.4871, 6.996, 2.3, 1.5, 4.3, 107, 0.53),
('2019-06-17', '22:02:52.389', 50.3840, 7.5295, 4.318, 3.8, 1.9, 4.1, 183, 0.18),
('2019-06-25', '21:18:03.029', 50.3908, 7.5240, 6.376, 3.2, 1.7, 3.9, 82, 0.25),
('2019-07-19', '00:56:03.274', 50.3965, 7.5109, 7.986, 4.3, 2.0, 2.7, 113, 0.13),
('2019-07-21', '16:43:51.113', 50.3754, 7.2965, 27.214, 2.2, 1.7, 3.3, 54, 0.24),
('2019-08-07', '21:13:39.219', 50.3837, 7.2954, 38.050, 2.7, 3.6, 3.0, 59, 0.17),
('2019-10-29', '17:43:52.984', 50.3755, 7.3117, 39.450, 3.1, 3.0, 4.6, 49, 0.22),
('2020-02-21', '18:27:04.950', 50.3976, 7.4763, 5.555, 2.3, 1.5, 4.6, 81, 0.20),
('2020-03-09', '21:08:13.319', 50.3938, 7.4844, 9.681, 2.3, 2.4, 4.6, 152, 0.33),
('2021-04-07', '22:28:43.340', 50.3707, 7.3167, 35.681, 1.8, 1.8, 2.7, 41, 0.12),
('2021-05-06', '21:59:47.373', 50.3737, 7.3088, 37.250, 2.1, 2.2, 2.6, 39, 0.19),
('2021-05-10', '18:39:00.484', 50.3664, 7.3117, 35.192, 2.2, 1.8, 3.3, 43, 0.16),
('2021-06-19', '19:09:10.298', 50.3667, 7.3197, 36.191, 3.0, 2.1, 4.0, 44, 0.17),
('2021-08-22', '01:18:35.463', 50.3619, 7.3184, 34.417, 2.7, 2.8, 4.3, 51, 0.15),
('2021-11-17', '02:16:54.154', 50.3655, 7.3178, 36.084, 2.9, 2.5, 4.0, 73, 0.09),
('2021-11-17', '02:17:25.057', 50.3623, 7.2940, 36.329, 3.5, 2.6, 4.2, 81, 0.15),
('2021-11-17', '02:42:04.710', 50.3658, 7.2839, 36.183, 3.0, 2.4, 4.4, 81, 0.26),
('2021-11-17', '02:52:07.584', 50.3687, 7.3117, 37.017, 3.2, 2.6, 3.5, 73, 0.18),
('2021-11-17', '02:53:16.302', 50.3692, 7.3100, 35.257, 3.0, 2.8, 4.3, 72, 0.14),
('2021-11-20', '02:57:39.857', 50.3675, 7.3146, 35.772, 3.4, 2.8, 4.3, 73, 0.14)
go

-- select * from t_events


insert into t_eventsQuality(evQua,nrPPic,nrSPic) values
('A',4,8),
('A',7,12),
('A',5,11),
('A',0,7),
('A',4,6),
('A',6,13),
('A',3,11),
('A',10,10),
('A',2,5),
('A',6,10),
('A',10,12),
```

```
('A',9,11),
('A',0,13),
('A',5,10),
('A',2,5),
('A',3,12),
('A',1,13),
('A',6,12),
('A',8,9),
('A',7,13),
('B',9,11),
('B',3,12),
('B',4,5),
('B',6,6),
('B',1,7),
('B',2,8),
('B',10,14),
('B',8,10),
('B',5,11),
('B',4,13),
('B',7,8),
('B',3,7),
('B',0,6),
('B',10,15),
('B',2,9),
('B',1,8),
('B',9,12),
('B',6,9),
('B',3,6),
('B',8,10),
('C',5,13),
('C',4,10),
('C',7,9),
('C',10,12),
('C',6,11),
('C',0,8),
('C',2,6),
('C',9,12),
('C',1,7),
('C',8,9)
GO
-- select * from t_eventsQuality

insert into
t_stations(stCod,stNet,Lat,Lon,alti,eigenFrq,onLin,tempoPerm,instalDate,agencName
) values
('DE01','KB',50.3927,7.2069,123,1,'off','permanent','2013-05-01','KIT'),
('DE02','KB',50.4181,7.3181,100,1,'off','temporary','2015-07-01','KIT'),
('DE03','KB',50.3551,7.9935,153,1,'off','temporary','2015-07-01','KIT'),
('DE04','KB',50.1509,7.0550,110,1,'off','temporary','2015-07-02','KIT'),
('DE05','KB',50.3607,7.1025,133,round(1.0/120,4),'off','temporary','2015-07-02','KIT'),
('DE06','KB',49.7017,7.0590,160,1,'off','temporary','2015-07-02','KIT'),
('DE07','KB',50.1556,7.5312,123,1,'off','temporary','2015-07-03','KIT'),
('DE08','KB',50.2370,7.2897,100,1,'off','temporary','2015-07-03','KIT'),
('DE09','KB',50.3754,7.2842,113,1,'off','temporary','2015-07-03','KIT'),
('DE10','KB',50.4208,7.5865,145,1,'on','temporary','2016-03-01','KIT'),
('DE11','KB',50.5043,7.5774,115,1,'off','temporary','2016-03-01','KIT'),
('DE12','KB',50.5447,7.4337,160,round(1.0/120,4),'on','permanent','2013-02-01','KIT'),
('DE13','KB',50.5842,7.2414,140,1,'off','temporary','2016-04-01','KIT'),
('ABH ','LE',49.8820,7.5480,132,round(1.0/120,4),'off','permanent','2014-10-01','Erdbebendienst Südwest'),
('OCHT','LE',50.3387,7.3575,111,1,'off','permanent','2014-10-01','Erdbebendienst Südwest'),
('GLOK','LE',50.5447,7.4337,190,round(1.0/120,4),'off','permanent','2017-05-21','Erdbebendienst Südwest'),
('NICK','GR',50.4180,7.3180,150,round(1.0/120,4),'on','permanent','2017-05-21','German Regional Seismic Network')
GO
-- select * from t_stations

insert into
t_stInstalService(stCod,lastDateServ,personFulName1,personFulName2,sensorTyp,GPSS
tatus,nrGPSSat) values
('DE01','2016-02-01','Konun Koushesh','Joachim Ritter','Mark-3D','on',6),
('DE02','2016-02-01','Konun Koushesh','Joachim Ritter','Mark-3D','on',3),
('DE03','2016-02-01','Konun Koushesh','Joachim Ritter','Mark-3D','on',5),
```

```
('DE04','2016-02-01','Konun Koushesh','Joachim Ritter','Mark-3D','on',6),
('DE05','2017-03-11','Werner Schärer','Joachim Ritter','STS2','on',2),
('DE06','2017-03-11','Werner Schärer','Joachim Ritter','Mark-3D','on',9),
('DE07','2017-03-11','Werner Schärer','Joachim Ritter','Mark-3D','on',7),
('DE08','2017-03-11','Werner Schärer','Joachim Ritter','Mark-3D','on',9),
('DE09','2017-03-12','Werner Schärer','Joachim Ritter','Mark-3D','on',4),
('DE10','2017-03-12','Werner Schärer','Joachim Ritter','Mark-3D','on',3),
('DE11','2018-09-01','Werner Schärer','Konun Koushesh','Mark-3D','on',6),
('DE12','2018-09-01','Werner Schärer','Konun Koushesh','STS2','on',8),
('DE13','2018-09-01','Werner Schärer','Konun Koushesh','Mark-3D','on',6),
('ABH ','2018-09-02','Werner Schärer','Konun Koushesh','STS2','on',10),
('OCHT','2018-09-02','Werner Schärer','Konun Koushesh','Mark-3D','on',2),
('GLOK','2018-09-02','Werner Schärer','Konun Koushesh','STS2','on',5),
('NICK','2018-09-02','Werner Schärer','Konun Koushesh','STS2','on',7)
GO

update t_stInstalService set issues = 'there is regular noise in the records'
where stid =5
-- select * from t_stInstalService


insert into
t_ownerStationSite(personFulName1,personFulName2,email,fullAdress,tel) values
('Anna Schuster','Klaus Schuster','anna.schuster@example.de','Hauptstrasse 12, 56073 Kobenz','+49 151 23456701'),
('Jonas Müller','Nick Mueller','jonas.mueller@example.de','Bergstraße 45, 56073 Kobenz','+49 157 76543210'),
('Mia Schröder','Ellena Schroeder','mia.schroeder@example.de','Bahnhofstraße 7,  56073 Kobenz','+49 160 98765432'),
('Paul Fischer','Anna Fischer','paul.fischer@example.de','Lindenweg 23,  56073 Kobenz','+49 170 11223344'),
('Sophie Weber','Dirk Weber','sophie.weber@example.de','Marktplatz 5,  56073 Kobenz','+49 152 99887766'),
('Tim Wagner','Timo Wagner','tim.wagner@example.de','Friedrichstraße 8, 56743 Mendig','+49 151 33445566'),
('Lea Becker','Toni Becker','lea.becker@example.de','Poststraße 19,  56073 Kobenz','+49 157 55667788'),
('Max Braun','Peter Braun','max.braun@example.de','Am Ring 3,  56073 Kobenz','+49 160 22334455'),
('Emma Hoffmann','Sandra Hoffmann','emma.hoffmann@example.de','Rosenweg 10,  56073 Kobenz','+49 170 44332211'),
('Noah Koch','Frank Koch','noah.koch@example.de','Schillerstraße 33, 56743 Mendig','+49 152 66778899'),
('Julia Richter','Dario Richter','julia.richter@example.de','Mozartstraße 12,  56073 Kobenz','+49 151 10101010'),
('Tom Schäfer','Jimi Shaefer','tom.schaefer@example.de','Waldweg 8,  56073 Kobenz','+49 157 90909090'),
('Alina Vogel','Landao Vogel','alina.vogel@example.de','Bachstraße 27,  56073 Kobenz','+49 160 30303030'),
('Jan Klein','Fred Klein','jan.klein@example.de','Goethestraße 4,  56073 Kobenz','+49 170 40404040'),
('Clara Hartmann','Karl Hartmann','clara.hartmann@example.de','Allee 16,  56073 Kobenz','+49 152 50505050'),
('Felix Schröder','Bernhard Schroeder','felix.schroeder@example.de','Hafenstraße 2,  56073 Kobenz','+49 151 60606060'),
('Laura Böhm','Nava Boehm','laura.boehm@example.de','Ringstraße 18, 56743 Mendig','+49 157 70707070')
GO

update t_ownerStationSite set issues = 'available only on between 08:00 and
12:00' where stid =17
-- select * from t_ownerStationSite


insert into t_remarkablePlaces(placName,Lat,Lon) values
('Laacher See volcano',50.413259,7.271717),
('Magmatic activity epicenter main',50.375246, 7.317210),
('Magmatic activity epicenter east',50.428246, 7.409210),
('Glees',50.4333,7.2433),
('Maria Laach',50.4022,7.2519),
('Nickenich',50.4500,7.2333),
('Kruft',50.4169,7.2950),
('Wassenach',50.4267,7.2692),
('Mendig',50.3833,7.3000),
('Kottenheim',50.3500,7.2000),
('Mayen',50.4333,7.2700),
('Andernach',50.4469,7.3928),
('Bad Neuenahr-Ahrweiler',50.5400,7.1178),
('Oberzissen',50.3900,7.2750),
('Ettringen',50.3778,7.3167),
('Plaidt',50.4075,7.3500),
('Burgbrohl',50.4667,7.2833),
('Wehr',50.4500,7.2333),
('Niederzissen',50.3833,7.2833),
('Bell',50.4333,7.2333),
('Koblenz',50.3569,7.5880),
('Wiesbaden',50.0826,8.2408),
('Bonn', 50.7358509993, 7.10066),
('Cologne', 50.933594, 6.961899),
```

```
('Frankfurt', 50.110924, 8.682127),
('Düsseldorf', 51.233334, 6.783333),
('Essen', 51.450832, 7.013056),
('Dortmund', 51.514244, 7.468429),
('Bochum', 51.4818111, 7.2196635),
('Aachen', 50.7763509993, 6.0838619999),
('Mannheim', 49.488888, 8.469167),
('Karlsruhe', 49.006889, 8.403653),
('Bielefeld', 52.0191005, 8.531007),
('Gelsenkirchen', 51.516666, 7.1),
('Wuppertal', 51.256214, 7.150764),
('Leipzig', 51.340199, 12.360103),
('Dresden', 51.050407, 13.737262),
('Munich', 48.137154, 11.576124),
('Stuttgart', 48.783333, 9.183333),
('Nuremberg', 49.460983, 11.061859),
('Augsburg', 48.3668041, 10.8986971),
('Hamburg', 53.551086, 9.993682),
('Berlin', 52.520008, 13.404954),
('Bremen', 53.073635, 8.806422),
('Kassel', 51.312801, 9.481544),
('Magdeburg', 52.133331, 11.616667),
('Halle (Saale)', 51.483334, 11.966667),
('Mainz', 49.992863, 8.247253),
('Heidelberg', 49.39875, 8.672434),
('Trier', 49.75, 6.633333),
('Brussels', 50.8503463, 4.3517211),
('Ghent', 51.0543422, 3.7174246),
('Antwerp', 51.2194475, 4.4024643),
('Rotterdam', 51.926517, 4.462456),
('The Hague', 52.078663, 4.288788),
('Amsterdam', 52.371807, 4.896029),
('Utrecht', 52.092876, 5.104480),
('Gouda', 52.011112, 4.711111),
('Leiden', 52.160114, 4.497010),
('Haarlem', 52.387386, 4.646219),
('Luxembourg', 49.611622, 6.131934),
('Liège', 50.632557, 5.579666),
('Namur', 50.467387, 4.871987),
('Metz', 49.119308, 6.175715),
('Strasbourg', 48.5734053, 7.7521113),
('Nancy', 48.692054, 6.184417),
('Lyon', 45.764043, 4.835659),
('Paris', 48.856614, 2.3522219),
('Dijon', 47.322047, 5.04148),
('Liège', 50.632557, 5.579666),
('Rouen', 49.443232, 1.099971),
('Reims', 49.258329, 4.031696),
('Lille', 50.62925, 3.057256),
('Toulouse', 43.604652, 1.444209),
('Nice', 43.710173, 7.261953),
('Marseille', 43.296482, 5.36978),
('Bruges', 51.209348, 3.2247),
('Liège', 50.632557, 5.579666),
('Namur', 50.467387, 4.871987),
('Luxembourg City', 49.611622, 6.131934),
('Bendorf',50.426268, 7.562109)
GO

-- select * from t_remarkablePlaces
```

3. Create Procedure

```sql
-- a procedure to provide call/contact info of the owner person of the site that
a station there is installed
--> one must give a station ID while executing the procedure
CREATE OR ALTER PROCEDURE sp_join_stations_ownerStationSite
    @stid INT,
    @stCod CHAR(4) OUTPUT,
    @Lat float OUTPUT,
    @Lon float OUTPUT,
    @personNam1 NVARCHAR(50) OUTPUT,
    @personNam2 NVARCHAR(50) OUTPUT,
    @tel NVARCHAR(30) OUTPUT,
    @adress NVARCHAR(50) OUTPUT,
    @issues NVARCHAR(50) OUTPUT
AS
BEGIN
    SELECT
        @stCod = st.stCod,
        @Lat = st.Lat,
      @Lon = st.Lon,
        @personNam1 = ownSt.personFulName1,
        @personNam2 = ownSt.personFulName2,
        @tel = ownSt.tel,
        @adress = ownSt.fullAdress,
      @issues = ownSt.issues
    FROM t_stations AS st
    JOIN t_ownerStationSite AS ownSt ON st.stid = ownSt.stid
    WHERE st.stid = @stid;
END;
GO
-- execute the procedure
DECLARE
    @stCod CHAR(4),
    @Lat float,
    @Lon float,
    @personNam1 NVARCHAR(50),
    @personNam2 NVARCHAR(50),
    @tel NVARCHAR(30),
    @adress NVARCHAR(50),
    @issues NVARCHAR(50);

-- give the station ID (stid) to see the corresponding following declared query
results
EXEC sp_join_stations_ownerStationSite
    @stid = 1,  -- Replace 1 with the station ID you want
    @stCod = @stCod OUTPUT,
    @lat = @lat OUTPUT,
    @lon = @lon OUTPUT,
    @personNam1 = @personNam1 OUTPUT,
    @personNam2 = @personNam2 OUTPUT,
    @tel = @tel OUTPUT,
    @adress = @adress OUTPUT,
    @issues = @issues OUTPUT;
SELECT
    @stCod AS stCod,
    @lat as Lat,
    @lon as Lon,
    @personNam1 AS first_contact_person,
    @personNam2 AS second_contact_person,
    @tel AS tel,
    @adress AS adress,
    @issues as availiblity;
```

4. Creating Trigger

```sql
 -- create a table in which changes like insert, delete and update in the table
t_stInstalService are inserted to it
CREATE TABLE t_stInstalService_delet_update(
      stid int IDENTITY(1,1) PRIMARY KEY NOT NULL,
      stCod char(4) NULL,
      lastDateServ date NULL,
      personFulName1 nvarchar(50) NULL,
      personFulName2 nvarchar(50) NULL,
      sensorTyp nvarchar(30) null,
      GPSStatus nvarchar(3) null,
      nrGPSSat int null,
      issues nvarchar(50) default 'no issues'
)
GO
-- create a trigger to insert changes of the table t_stInstalService into
t_stInstalService_delet_update
CREATE OR ALTER TRIGGER dbo.tr_stInstalService_delet_update
ON dbo.t_stInstalService
AFTER INSERT, DELETE, UPDATE
AS
BEGIN
    SET NOCOUNT ON;
    INSERT INTO t_stInstalService_delet_update (
        stCod,
        lastDateServ,
        personFulName1,
        personFulName2,
        sensorTyp,
        GPSStatus,
        nrGPSSat,
        issues
    )
    SELECT
        stCod,
        lastDateServ,
        personFulName1,
        personFulName2,
        sensorTyp,
        GPSStatus,
        nrGPSSat,
        issues
    FROM deleted;
END
GO
--  test if trigger works correct after altering the table t_stInstalService
using update, delete and insert
SELECT * FROM t_stInstalService;

UPDATE t_stInstalService SET lastDateServ = '2020-01-12' WHERE stid = 1;
UPDATE t_stInstalService SET lastDateServ = '2020-01-13' WHERE stid = 4;
UPDATE t_stInstalService SET lastDateServ = '2022-05-20' WHERE stid = 8;
UPDATE t_stInstalService SET lastDateServ = '2024-08-02' WHERE stid = 12;
DELETE FROM t_stInstalService WHERE stid = 13;

SET IDENTITY_INSERT t_stInstalService on;
insert into
t_stInstalService(stid,stCod,lastDateServ,personFulName1,personFulName2,sensorTyp
,GPSStatus,nrGPSSat,issues)
values(13,'DE13','2025-03-12','Joachim Ritter','Konun
Koushesh','STS2','on',6,'the sensore is switched from Mark-3D to STS2');
SET IDENTITY_INSERT t_stInstalService off;

SELECT * FROM t_stInstalService_delet_update;
SELECT * FROM t_stInstalService;
```

```sql
-- to disable or again enable the trigger use unmute the one of the followings:

--DISABLE TRIGGER dbo.tr_stInstalService_delet_update
--ON dbo.t_stInstalService
--GO

--ENABLE TRIGGER dbo.tr_stInstalService_delet_update
--ON dbo.t_stInstalService
--GO
```

5. Create Index

```sql
-- index to speed up the search (filter) criterion
CREATE NONCLUSTERED INDEX idx_events ON t_events(Lon);
SELECT * FROM t_events WHERE Lon > 7.35;

-- some queries and updates  on tables --

-- in t_events or t_eventsQuality there is already known that two cluster of
events exist:
-- one include more activities (the main cluster), another one has less activity
(the east cluster)
-- the separation border for these clusters is the value of 7.35 as the longitude

-- in the following this criterion is used to separate these clusters
-- and also calculate the average of the locations of events in each cluster

-- at the end of this part these averaged values are inserted
-- to their corresponding rows in the table t_remarkablePlaces

-- declaration of tow sets of variables:
DECLARE @latEast FLOAT, @lonEast FLOAT, @depthEast FLOAT;
DECLARE @latMain FLOAT, @lonMain FLOAT, @depthMain FLOAT;
-- applying the filter for:
-- the east cluster
SELECT
    @latEast = AVG(Lat),
    @lonEast = AVG(Lon),
    @depthEast = AVG(depth)
FROM t_events
WHERE Lon > 7.35;
-- for the main cluster
SELECT
    @latMain = AVG(Lat),
    @lonMain = AVG(Lon),
    @depthMain = AVG(depth)
FROM t_events
WHERE Lon <= 7.35;
-- Print the result
PRINT 'East cluster activity center: Lat: '
    + CHAR(13) + CHAR(10) -- to go to the next line
    + LTRIM(STR(@latEast, 10, 4))
    + ' Lon: ' + LTRIM(STR(@lonEast, 10, 4))
    + ' Depth: ' + LTRIM(STR(@depthEast, 10, 2));
PRINT 'Main cluster activity center: Lat: '
     + CHAR(13) + CHAR(10) -- to go to the next line
     + LTRIM(STR(@latMain, 10, 4))
     + ' Lon: ' + LTRIM(STR(@lonMain, 10, 4))
     + ' Depth: ' + LTRIM(STR(@depthMain, 10, 2));

update t_remarkablePlaces set Lat = 50.3748, Lon = 7.3032 where placid = 2;
update t_remarkablePlaces set Lat = 50.3928, Lon = 7.5014 where placid = 3;

-- drop index idx_events on t_events;
```

6. Create Funktion

```
-- generating a function to sort (order) places by distance to an assumed
eruption point in the east Eifel volcanic field

-- function which takes an place ID and returns the latitude of the place
CREATE OR ALTER FUNCTION takePlacIdReturnPlaceLat
(@placid INT)
RETURNS float
AS
BEGIN
    DECLARE @latEruptionPoint float;
    SELECT @latEruptionPoint = Lat
    FROM t_remarkablePlaces
    WHERE placid = @placid;
    RETURN @latEruptionPoint
END;
GO
-- function which takes an place ID and returns the longitude of the place
CREATE OR ALTER FUNCTION takePlacIdReturnPlaceLon
(@placid INT)
RETURNS float
AS
BEGIN
    DECLARE @lonEruptionPoint float;
    SELECT  @lonEruptionPoint = Lon
    FROM t_remarkablePlaces
    WHERE placid = @placid;
    RETURN @lonEruptionPoint
END;
GO
-- function which calculates the distances between each place in the table and
the assumed eruption point (input)
CREATE OR ALTER FUNCTION nearestPlacesToEruptionPoint
(@latEruptionPoint float, @lonEruptionPoint float)
RETURNS TABLE
AS
RETURN
(SELECT
     placid,placName,

ROUND(geography::Point(lat,lon,4326).STDistance(geography::Point(@latEruptionPoin
t,@lonEruptionPoint,4326)) / 1000.0, 2) AS DistanceToEruptionPoint_km
FROM t_remarkablePlaces);
GO

-- a nesting function to see the places sorted by distance to the assumed
eruption point (e.g.: place ID = 1)
-- give a place ID: 1, 2 or 3 to see the nearest places to it
-- place ID = 1 = Laacher See volcano
-- place ID = 2 = Magmatic activity epicenter main
-- place ID = 3 = Magmatic activity epicenter east

-- eruption center 1
DECLARE @lat FLOAT = dbo.takePlacIdReturnPlaceLat(1);
DECLARE @lon FLOAT = dbo.takePlacIdReturnPlaceLon(1);
INSERT INTO t_remarkablePlacesCloseToeruptionCenter1 (placid, placName,
distanceToEruptionPoint_km)
SELECT placid, placName, DistanceToEruptionPoint_km
FROM dbo.nearestPlacesToEruptionPoint(@lat, @lon);

-- eruption center 2
DECLARE @lat FLOAT = dbo.takePlacIdReturnPlaceLat(2);
DECLARE @lon FLOAT = dbo.takePlacIdReturnPlaceLon(2);
INSERT INTO t_remarkablePlacesCloseToeruptionCenter2 (placid, placName,
distanceToEruptionPoint_km)
```

```sql
SELECT placid, placName, DistanceToEruptionPoint_km
FROM dbo.nearestPlacesToEruptionPoint(@lat, @lon);

-- eruption center 3
DECLARE @lat FLOAT = dbo.takePlacIdReturnPlaceLat(3);
DECLARE @lon FLOAT = dbo.takePlacIdReturnPlaceLon(3);
INSERT INTO t_remarkablePlacesCloseToeruptionCenter3 (placid, placName,
distanceToEruptionPoint_km)
SELECT placid, placName, DistanceToEruptionPoint_km
FROM dbo.nearestPlacesToEruptionPoint(@lat, @lon);

-- View
CREATE OR ALTER VIEW v_remarkablePlacesCloseToEruptionCenter123 AS
SELECT
    center1.placid,
    center1.placName AS placeName,
    center1.distanceToEruptionPoint_km AS distanceToCenter1_km,
    center2.distanceToEruptionPoint_km AS distanceToCenter2_km,
    center3.distanceToEruptionPoint_km AS distanceToCenter3_km
FROM t_remarkablePlacesCloseToeruptionCenter1 center1
JOIN t_remarkablePlacesCloseToeruptionCenter2 center2 ON center1.placid =
center2.placid
JOIN t_remarkablePlacesCloseToeruptionCenter3 center3 ON center1.placid =
center3.placid;

SELECT * FROM v_remarkablePlacesCloseToEruptionCenter123
--order by distanceToCenter1_km , distanceToCenter2_km, distanceToCenter3_km
```

8. Create View

```sql
-- create simple views first
create view v_events
as
select * from t_events;

SELECT * FROM v_events;
--
create view v_eventsQuality
as
select * from t_eventsQuality;

SELECT * FROM v_eventsQuality;
--
create view v_stations
as
select * from t_stations;

SELECT * FROM v_stations;
--
create view v_stInstalService
as
select * from t_stInstalService;

SELECT * FROM v_stInstalService;
--
create view v_ownerStationSite
as
select * from t_ownerStationSite;

SELECT * FROM v_ownerStationSite;
--
create view v_remarkablePlaces
as
select * from t_remarkablePlaces;

SELECT * FROM v_remarkablePlaces;
```

```sql
-- create a view of merging data of the two tables t_events and t_eventsQuality
create view v_join_events_eventsQuality
as
select
    ev.evid,
    ev.date,
    ev.time,
    ev.Lat,
    ev.Lon,
    ev.depth,
    ev.latEr,
    ev.lonEr,
    ev.depEr,
    ev.angGap,
    ev.picEr,
    evQu.evQua,
    evQu.nrPPic,
    evQu.nrSPic
from t_events ev
join t_eventsQuality evQu on ev.evid = evQu.evid

SELECT * FROM v_join_events_eventsQuality

-- create a view of merging data of the three tables t_stations,
t_stInstalService and t_ownerStationSite
create view v_join_stations_ownerStationSite_stInstalService
as
select
     st.stid,
     st.stCod,
     st.stNet,
     st.Lat,
     st.Lon,
     st.alti,
     st.eigenFrq,
     st.onLin,
     st.tempoPerm,
     st.instalDate,
     st.agencName,
     inst.lastDateServ,
     inst.personFulName1 as instalServic_person1,
     inst.personFulName2 as instalServic_person2,
     inst.sensorTyp,
     inst.GPSStatus,
     inst.nrGPSSat,
     inst.issues as issues_on_instalService,
     own.personFulName1 as owner_person1,
     own.personFulName2 as owner_person2,
     own.email,
     own.tel,
     own.fullAdress,
     own.issues as accessablity
from t_stations st
join t_stInstalService inst on st.stid = inst.stid
join t_ownerStationSite own on st.stid = own.stid

SELECT * FROM v_join_stations_ownerStationSite_stInstalService


-- there is a view in the script seismoSQL_createFunction which shows
-- the distances between three assumed eruption centers in
-- the east Eifel volcanic field and some known villages/cities up to 1000 km
radius
```